

FRAUD MONITORING & AML KNOWLEDGE BASE

◆ 1. Introduction to Fraud Detection

Fraud detection in banking systems involves identifying suspicious or unauthorized transactions using rule-based and behavior-based mechanisms. Financial institutions use fraud monitoring systems to prevent financial loss, ensure compliance, and protect customer accounts.

Fraud detection systems typically analyze:

- Transaction patterns
 - User behavior
 - Geographical anomalies
 - Transaction frequency and velocity
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◆ 2. Types of Fraud in Banking

2.1 Transaction Fraud

Unauthorized or suspicious transfer of funds between accounts.

2.2 Account Takeover (ATO)

When an attacker gains access to a user's account and performs fraudulent transactions.

2.3 Card Fraud

Fraudulent use of debit/credit card details for unauthorized payments.

2.4 Identity Theft

Using stolen identity information to create fake accounts or perform transactions.

2.5 Phishing Attacks

Fraudulent attempts to obtain sensitive data like passwords or OTPs.

◆ 3. Fraud Classification

Transactions are classified into three categories:

- **VALID:** Normal transactions with no suspicious behavior
 - **ALERT:** Suspicious transactions requiring further review
 - **FRAUD:** High-risk transactions confirmed as fraudulent
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◆ 4. Fraud Detection Rules

4.1 High Amount Rule

Transactions exceeding a predefined threshold (e.g., ₹1,00,000) are flagged.

4.2 Velocity Rule

Multiple transactions within a short time period indicate suspicious activity.

4.3 Geolocation Rule

Transactions from unusual or distant locations within a short time.

4.4 New Beneficiary Rule

Transactions to newly added accounts are considered risky.

4.5 Odd Timing Rule

Transactions during unusual hours (e.g., midnight) may indicate fraud.

◆ 5. Fraud Risk Scoring

Each transaction is assigned a risk score (0–100) based on:

- Transaction amount
- Frequency
- Location mismatch
- Historical behavior
- Device/IP anomalies

Risk Levels:

- 0–30 → Low Risk

- 31–70 → Medium Risk (ALERT)
 - 71–100 → High Risk (FRAUD)
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◆ 6. AML (Anti-Money Laundering) Guidelines

AML policies ensure financial institutions detect and prevent money laundering.

Key AML Principles:

- Customer Due Diligence (CDD)
 - Know Your Customer (KYC)
 - Transaction Monitoring
 - Suspicious Activity Reporting (SAR)
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◆ 7. Suspicious Transaction Indicators

A transaction may be suspicious if:

- Large amount transfers occur suddenly
 - Repeated failed transactions occur
 - Transactions to unknown accounts
 - Rapid fund movement across accounts
 - Frequent international transfers
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◆ 8. Fraud Investigation Process

1. Detect suspicious transaction
 2. Assign risk score
 3. Flag as ALERT or FRAUD
 4. Analyst reviews transaction
 5. Account may be blocked
 6. Report generated
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◆ 9. Account Blocking Rules

Accounts may be blocked if:

- Multiple fraud transactions detected

- High-risk score exceeded
 - Suspicious login patterns observed
 - AML violations detected
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◆ 10. Fraud Patterns

Common fraud patterns include:

- Rapid transactions (velocity fraud)
 - Round amount transactions (₹100000, ₹50000)
 - Repeated failed attempts
 - Cross-border anomalies
 - Same sender → multiple receivers
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◆ 11. Fraud Prevention Techniques

- Multi-factor authentication (MFA)
 - Real-time transaction monitoring
 - AI-based anomaly detection
 - Behavioral biometrics
 - Device fingerprinting
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◆ 12. Analyst Guidelines

Fraud analysts should:

- Review flagged transactions
 - Verify customer identity
 - Analyze transaction history
 - Take action (block / monitor)
 - Document findings
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◆ 13. Compliance & Reporting

- Suspicious transactions must be reported
- Maintain audit logs
- Follow regulatory guidelines (RBI, FATF)

◆ 14. Key Terms

- **Fraud:** Unauthorized financial activity
 - **AML:** Anti-Money Laundering
 - **KYC:** Know Your Customer
 - **Risk Score:** Fraud probability measure
 - **Alert:** Suspicious but not confirmed fraud
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